

DocuSign Clone — Comprehensive QA Audit Summary

Scaler AI Labs · RL Environments Quality Assurance Pass (38 Quality Bugs)

Target Viewport: 1280 × 960 (100% Zoom)

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1. OVERALL PRODUCT IMPRESSION

The clone replicates DocuSign's macro design language, navigation shell, and core envelope layout. However, extensive functional auditing revealed **38 high-impact defects** heavily concentrated in backend state invariants, signer session stability, canvas tag coordinate scaling, and task grader consistency. In an RL agent evaluation pipeline (e.g., Anthropic Computer Use or OpenAI Operator), these flaws would introduce severe false-positive grader passes, trajectory crashes, and unrecoverable task loops.

2. TOP 5 HIGH-IMPACT BUGS TO FIX FIRST (RL & ENTERPRISE IMPACT)

- P0 BUG-016 (State Invariant Failure: Empty Envelope PUT to 'Sent'):** Direct API mutation allows transitioning envelopes with 0 documents and 0 recipients to 'sent' status. In RL training, autonomous models exploit this loophole to falsely achieve reward criteria without performing the requisite workflow actions.
- P0 BUG-017 (White-Screen Runtime Crash on Fieldless Signing Session):** Accessing `/sign/[envelopeId]` when an envelope contains no pre-placed fields causes an unhandled React `TypeError`, completely blocking human and agent signers.
- P1 BUG-001 & BUG-020 (Send Flow & Recipient Role Validation Bypasses):** Zero-recipient submissions are permitted through the prepare UI, while read-only CC recipients incorrectly block dispatch unless a signature tag is placed.
- P1 BUG-002 & BUG-037 (Canvas Tag Volatility & Zoom Transform Drift):** Placed signature tags vanish on browser reload, and canvas zoom controls scale CSS visually without recalculating relative coordinate matrices, causing 60px drop offsets.
- P1 BUG-024 (Filter Query Failure on Status Views):** Querying agreements with `?status=voided` or `?status=declined` ignores parameters and returns active 'sent' envelopes, invalidating grader assertions on envelope lifecycle state.

3. TESTING SCOPE & METHODOLOGY

In-Scope Audited (38 Bugs): HTTP Basic Auth session lifecycle, Dashboard metrics & quick actions, Multi-document upload & thumbnail rail, Recipient CRUD & sequential routing, PDF canvas drag-and-drop & tag persistence, Signer ceremony (`/sign/[id]`) & Adopt Signature pad, Template application, Agreement list filters & detail views, and DevTools console/CSS computed styles.

Deliberately Skipped (Out-of-Scope): File exports (PDF/CSV generation), third-party integrations (Drive, Slack), and penetration/brute-force auth testing per assignment boundaries.

4. ASSUMPTIONS & UNCONFIRMED SUSPICIONS

- Assumption:** The clone operates as a single-tenant mocked environment where asynchronous email notification dispatch is simulated in-memory without persistent message broker queues.
- Unconfirmed Suspicion:** Canvas drag-and-drop bounding boxes lack dynamic viewport resizing listeners, causing field alignment to drift significantly on non-standard document aspect ratios.